

On k -stars in perfect binary trees

Abstract

We prove that for every perfect binary tree T_h of height $h \geq 0$ and every integer $k \geq 1$, the number of independent sets of size k containing a given vertex is maximized when that vertex is a leaf. Equivalently, for every vertex v of T_h and every leaf ℓ of T_h we have

$$s_k(v) \leq s_k(\ell),$$

where $s_k(w)$ denotes the number of independent sets of size k that contain the vertex w . The proof uses standard recursive (generating-function) decompositions of independent sets in rooted trees together with coefficientwise monotonicity arguments.

1 Notation and statement

Fix an integer $h \geq 0$. Let T_h denote the perfect rooted binary tree of height h : the root has depth 0, every internal vertex has exactly two children, and all leaves are at depth h . For a vertex $v \in V(T_h)$ and integer $k \geq 1$ define

$$s_k(v) := \#\{I \subseteq V(T_h) : I \text{ is independent, } |I| = k, v \in I\}.$$

We will write $S_{h,d}(k)$ for the common value of $s_k(v)$ for any vertex v at depth d (by symmetry all vertices at the same depth have the same k -star size). The main result is:

Theorem 1.1. *For every $h \geq 0$, every $k \geq 1$, and every vertex v of T_h ,*

$$s_k(v) \leq s_k(\ell)$$

for any leaf ℓ of T_h . Equivalently, the quantity $S_{h,d}(k)$ is nondecreasing in d for $0 \leq d \leq h$.

We prove the theorem by introducing generating functions that count independent sets in rooted perfect binary trees subject to whether the root is included or not, derive recurrences, and then compare the generating functions that arise when forcing a vertex at depth d into an independent set.

2 Generating functions for rooted perfect binary trees

For $t \geq 0$ let R_t denote the rooted perfect binary tree of height t (root at depth 0, leaves at depth t). Define two families of polynomials (ordinary generating functions)

$$A_t(x) := \sum_{m \geq 0} a_t(m)x^m, \quad B_t(x) := \sum_{m \geq 0} b_t(m)x^m,$$

where

- $a_t(m)$ is the number of independent sets of size m in R_t that *contain* the root (“root forced in”), and
- $b_t(m)$ is the number of independent sets of size m in R_t that *do not contain* the root (“root forced out”).

By convention A_t and B_t are polynomials (indeed finite) because R_t is finite. We also set

$$F_t(x) := A_t(x) + B_t(x),$$

the independence polynomial of R_t .

2.1 Base cases and recurrences

The tree R_0 is a single vertex (the root). Hence

$$A_0(x) = x, \quad B_0(x) = 1, \quad F_0(x) = 1 + x.$$

For $t \geq 1$ the root of R_t has two children, each of which is the root of a copy of R_{t-1} , and these two copies are disjoint. Standard combinatorial decomposition gives the recurrences:

$$A_t(x) = x \cdot B_{t-1}(x)^2, \quad B_t(x) = (A_{t-1}(x) + B_{t-1}(x))^2 = F_{t-1}(x)^2. \quad (1)$$

Intuitively: if the root is in the independent set then both child roots must be out, and the choices below each child are counted by B_{t-1} ; if the root is out then each child root may be in or out independently, so the choices at each child are counted by F_{t-1} and the two children contribute the square.

Combining these we also have

$$F_t(x) = A_t(x) + B_t(x) = xB_{t-1}(x)^2 + F_{t-1}(x)^2.$$

3 Expressing the k -star generating functions

Fix $h \geq 0$ and a depth $0 \leq d \leq h$. Let v be a vertex of T_h at depth d . We derive a factorization for the generating function

$$S_{h,d}(x) := \sum_{m \geq 0} S_{h,d}(m)x^m = \sum_{m \geq 0} s_m(v)x^m.$$

This polynomial enumerates independent sets of T_h that contain v by size.

If v is chosen to be in the independent set, its neighbors must be excluded. In the perfect binary tree this deletion separates the remaining vertices into disjoint rooted perfect binary subtrees (each such subtree is a copy of some R_t) and the sets of vertices chosen in each part are independent of each other. Hence the generating function factors as the product of the independence-generating polynomials of those parts, multiplied by x for the chosen vertex v itself.

We now describe the parts and their heights explicitly. Number the levels from the root (level 0) to the leaves (level h). Place v at level d . Then after removing v and its neighbors there are the following components:

1. The two subtrees rooted at the *grandchildren* of v (if they exist). Each such subtree is a copy of R_{h-d-2} (if $h-d-2 \geq 0$). If $h-d-2 < 0$ there are no such components and we interpret $F_t(x) = 1$ for $t < 0$.

2. The subtree rooted at the *sibling* of v (the other child of v 's parent), which is a copy of R_{h-d-1} (unless $d = 0$, in which case v has no sibling subtree counted this way).
3. For each ancestor of v strictly above its parent (there are $d - 1$ such ancestors when $d \geq 1$), at that ancestor the off-spine child (the child not on the path from the root to v) yields an entire copy of some R_t ; the other part of that level lies on the spine and is handled by the next ancestor. Concretely, for $i = 1, 2, \dots, d - 1$ the off-spine subtree encountered when moving from the parent of v upward i levels has height $h - d - 1 + i$.

Putting these factors together (and taking the empty product to be 1) yields the factorization

$$S_{h,d}(x) = x \cdot F_{h-d-2}(x)^2 \cdot (F_{h-d-1}(x))^{\mathbf{1}_{d \geq 1}} \cdot \prod_{i=1}^{d-1} F_{h-d-1+i}(x), \quad (2)$$

where $\mathbf{1}_{d \geq 1}$ is the indicator that equals 1 when $d \geq 1$ and 0 when $d = 0$. (The product over i is empty when $d \leq 1$.)

To see the formula more transparently: the root-ward part of the tree above the parent contributes one factor F_{h-d-1} (the sibling of v) and the $d - 1$ higher off-spine subtrees contribute the $\prod_{i=1}^{d-1} F_{h-d-1+i}$; below v the two grandchild subtrees each give F_{h-d-2} .

Remark 3.1. *When some indices are negative (for example when $d \geq h - 1$) we interpret $F_t(x) = 1$ for $t < 0$; this corresponds to the fact that certain components are absent.*

As a quick check, for a leaf ($d = h$) the formula becomes

$$S_{h,h}(x) = x \cdot F_0(x) \cdot \prod_{i=1}^{h-1} F_i(x) = x(1+x) \prod_{i=1}^{h-1} F_i(x),$$

since a leaf has a parent with a sibling subtree of height 0 and all higher off-spine subtrees of heights $1, 2, \dots, h - 1$.

For the root ($d = 0$) the formula becomes (empty sibling factor)

$$S_{h,0}(x) = x \cdot F_{h-2}(x)^2,$$

which matches the direct recurrence: choose the root (factor x) and below it there are two subtrees R_{h-1} whose roots must be out, so the choices for each are counted by $B_{h-1}(x) = F_{h-2}(x)$ (using (??) reversed). Indeed one checks that $B_t(x) = F_{t-1}(x)$ for $t \geq 1$ so the formulas are compatible.

4 Monotonicity in depth

To prove Theorem ?? it suffices to show that for each fixed h the polynomials $S_{h,d}(x)$ are coefficientwise nondecreasing in d as d increases from 0 to h . In other words we show

$$S_{h,d}(x) \preceq S_{h,d+1}(x) \quad (0 \leq d \leq h - 1),$$

where $P(x) \preceq Q(x)$ means that every coefficient of $Q - P$ is nonnegative.

We will use two simple monotonicity facts about the polynomials A_t, B_t, F_t .

Lemma 4.1. *For all $t \geq 0$ we have the coefficientwise inequalities*

- (a) $B_t(x) \preceq F_t(x)$,
- (b) $F_t(x) \preceq F_{t+1}(x)$.

Proof. (a) is immediate from $F_t = A_t + B_t$ (both A_t, B_t have nonnegative coefficients). For (b), observe from (??) that

$$F_{t+1}(x) = xB_t(x)^2 + F_t(x)^2 \succeq F_t(x)$$

coefficientwise because $xB_t(x)^2$ and $F_t(x)^2 - F_t(x)$ both have nonnegative coefficients (the latter because $F_t(x)^2 - F_t(x) = F_t(x)(F_t(x) - 1)$ and $F_t(x) - 1$ has nonnegative constant term when $t \geq 0$; more simply, every independent set of R_t is also an independent set of R_{t+1} by embedding, so $F_{t+1}(x) - F_t(x)$ counts new sets and has nonnegative coefficients). $\square$

Now compare $S_{h,d}(x)$ and $S_{h,d+1}(x)$ using the factorization (??). Write the two factorizations explicitly:

$$S_{h,d}(x) = x \cdot F_{h-d-2}(x)^2 \cdot F_{h-d-1}(x)^{\mathbf{1}_{d \geq 1}} \cdot \prod_{i=1}^{d-1} F_{h-d-1+i}(x),$$

$$S_{h,d+1}(x) = x \cdot F_{h-d-3}(x)^2 \cdot F_{h-d-2}(x)^{\mathbf{1}_{d+1 \geq 1}} \cdot \prod_{i=1}^d F_{h-d-2+i}(x).$$

Cancel the common factor x and compare the remaining multisets of F_* factors. One convenient way to view the relationship is as the result of the following replacements when d increases by 1:

- The factor $F_{h-d-1}(x)$ (which exists when $d \geq 1$) in $S_{h,d}$ is “replaced” by the factor $F_{h-d-2}(x)$ in $S_{h,d+1}$ (if $d = 0$ there was no such factor and $S_{h,1}$ simply gains a factor $F_{h-1-1} = F_{h-2}$ compared with $S_{h,0}$).
- The two factors $F_{h-d-2}(x)^2$ in $S_{h,d}$ are replaced by the two factors $F_{h-d-3}(x)^2$ in $S_{h,d+1}$.
- On the ancestor side, the product $\prod_{i=1}^{d-1} F_{h-d-1+i}(x)$ in $S_{h,d}$ becomes $\prod_{i=1}^d F_{h-d-2+i}(x)$ in $S_{h,d+1}$: that is, one factor at the lower end disappears while a factor with a strictly larger index appears at the upper end.

Using Lemma ??(b) we have $F_t(x) \preceq F_{t+1}(x)$ for every t , so each replacement described above is coefficientwise nondecreasing. More concretely, the product on the ancestor side is transformed by replacing (for each i) factors with smaller indices by factors with larger indices; coefficientwise monotonicity of F_t ensures the resulting product increases coefficientwise. Combining the three local monotonicity observations yields

$$S_{h,d}(x) \preceq S_{h,d+1}(x)$$

for all valid d , proving the desired monotonicity.

5 Conclusion and corollaries

The coefficientwise monotonicity $S_{h,d}(x) \preceq S_{h,d+1}(x)$ implies in particular that for every fixed $k \geq 1$ the coefficient of x^k satisfies

$$S_{h,d}(k) \leq S_{h,d+1}(k).$$

Iterating from $d = 0$ to $d = h$ yields

$$S_{h,0}(k) \leq S_{h,1}(k) \leq \cdots \leq S_{h,h}(k),$$

and since $S_{h,h}(k) = s_k(\ell)$ for any leaf ℓ we obtain Theorem ??.

Remark 5.1. *The proof above shows a stronger structural fact: the polynomial generating functions $S_{h,d}(x)$ are coefficientwise nondecreasing in the depth parameter d . In particular for each fixed k the numbers $S_{h,d}(k)$ form a nondecreasing sequence in d .*

Remark 5.2. *The same proof strategy applies to other regular rooted trees where the off-spine subtrees grow monotonically with depth; its key ingredients are that the independence polynomials for subtrees are coefficientwise monotone in their heights and that choosing a vertex separates the tree into independent components whose heights shift in a monotone way when the chosen vertex moves down the spine.*

Acknowledgments

The argument is standard in the combinatorics of independent sets in trees (see e.g. standard references on independence polynomials and combinatorial recurrences for trees). We wrote it here for completeness for the special case of perfect binary trees.